

Conjecture 00000000702 is false:
an algebraic p -adic number with periodic Browkin expansion

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1 The conjecture and its reading

Statement (English). *Definition: The Browkin expansion is a continued-fraction-like expansion in the p -adics. Conjecture: The Browkin expansion of an algebraic number is never eventually periodic; and the evidence for non-periodicity is the unboundedness of its partial quotients. (p -adic algebraic non-periodicity) The Chinese text says the same.*

Reading.

- p is an odd prime. Browkin's s -function (below) is defined for $p > 2$; Capuano–Murru–Terracini [CMT] write “we will fix a prime $p > 2$ ”. We refute the conjecture for *every* odd prime p .
- An *algebraic number* is an element $\alpha \in \mathbb{Q}_p$ that is algebraic over $\mathbb{Q}$. We restrict the claim to irrational α (for rational α the expansion is finite). This restricted claim is weaker, so refuting it refutes the original.
- Clause 1: the sequence of partial quotients is not eventually periodic. Clause 2: the partial quotients are unbounded. Browkin partial quotients always have real absolute value $< p/2$, so “unbounded” must refer to $|\cdot|_p$. We prove more: the partial quotients take only finitely many values, so they are bounded in every sense.
- “The Browkin expansion” is Browkin's (first) algorithm as stated in [CMT]. Because the Definition line is loose, we also treat Browkin's second algorithm, as stated in [MRS]. Each clause is refuted separately, for each algorithm.

Readings not refuted. (a) If “algebraic number” is meant to have degree ≥ 3 , then clause 1 becomes a known fact. [CMT] state: “It is easy to show that, if $\alpha \in \mathbb{Q}_p$ has periodic BCF expansion, then it is quadratic irrational”. Our quadratic example does not address clause 2 under that reading. (b) If clause 2 is read as the implication “non-periodic $\Rightarrow$ unbounded”, our periodic example is no counterexample to it. The conjunction is still false, because clause 1 fails.

2 Definitions

Let p be an odd prime. Let $\mathbb{Z}[1/p] = \{m/p^k : k \in \mathbb{N}, m \in \mathbb{Z}\} \subset \mathbb{Q}$.

Definition 1 ([CMT, Sec. 2]). Let $J_p = \mathbb{Z}[1/p] \cap (-p/2, p/2)$. [CMT] write: “given $\alpha \in \mathbb{Q}_p$ there exists a unique $s(\alpha) \in \mathcal{Y}$ such that $0 \leq |\alpha - s(\alpha)|_p < 1$ ” (their $\mathcal{Y}$ is our J_p). The *Browkin expansion* (algorithm (1) of [CMT]) is $\alpha_0 = \alpha$, $a_n = s(\alpha_n)$, $\alpha_{n+1} = 1/(\alpha_n - a_n)$ if $\alpha_n - a_n \neq 0$.

Definition 2 ([MRS, Sec. 1–2]). Write $\alpha = \sum_{n \geq -r} a_n p^n$ with digits $a_n \in \{-\frac{p-1}{2}, \dots, \frac{p-1}{2}\}$. Then $s(\alpha) = \sum_{n=-r}^0 a_n p^n$ and $t(\alpha) = \sum_{n=-r}^{-1} a_n p^n$. Browkin’s second algorithm (Browkin II, eq. (2) of [MRS]) is

$$b_n = \begin{cases} s(\alpha_n) & n \text{ even,} \\ t(\alpha_n) & n \text{ odd and } v_p(\alpha_n - t(\alpha_n)) = 0, \\ t(\alpha_n) - \text{sign}(t(\alpha_n)) & n \text{ odd and } v_p(\alpha_n - t(\alpha_n)) \neq 0, \end{cases} \quad \alpha_{n+1} = \frac{1}{\alpha_n - b_n}.$$

Characterisation of t . Let $K_p = \mathbb{Z}[1/p] \cap (-1/2, 1/2)$; this is the set K_p of [MRS]. By Lemma 2 of [MRS], $|t(\alpha)| < 1/2$. Also $\alpha - t(\alpha) = \sum_{n \geq 0} a_n p^n \in \mathbb{Z}_p$. So $t(\alpha)$ is an element $y \in K_p$ with $|\alpha - y|_p \leq 1$. Such a y is unique: if y, y' are two of them, then $y - y' \in \mathbb{Z}[1/p] \cap \mathbb{Z}_p = \mathbb{Z}$ and $|y - y'| < 1$. In the same way, $s(\alpha)$ is the unique $y \in J_p$ with $|\alpha - y|_p < 1$, in agreement with [CMT].

Lean encoding. `ZInvP p`, `browkinJ p` and `browkinK p` are the sets above. `browkinS x` is `Classical.epsilon` of the property “ $y \in J_p$ and $|x - y|_p < 1$ ”, and `browkinT x` is the same with K_p and $|x - y|_p \leq 1$. We prove uniqueness (`s_unique`, `t_unique`, valid for every prime). For odd p we prove existence for every $x \in \mathbb{Q}_p$ (`browkinS_spec`, `browkinT_spec`), using a balanced representative (`ZMod.valMinAbs`) of `PadicInt.appr`. So the Lean functions are exactly Browkin’s s and t .

- `bcfCQ/bcfPQ` are the complete and partial quotients of Definition 1. `bcf2CQ/bcf2PQ` (with `browkin2Digit`) are those of Definition 2.
- Conventions. The condition $v_p(x - t(x)) = 0$ is encoded as $|x - t(x)|_p = 1$. These are equivalent: for $x \neq t(x)$ this is the definition of v_p , and for $x = t(x)$ the valuation is $+\infty \neq 0$ while the norm is $0 \neq 1$. “sign” is Mathlib’s `SignType.sign` (± 1 , and 0 at 0). Lean’s convention $0^{-1} = 0$ never applies: we prove $\alpha_n \neq a_n$ (and $\alpha_n \neq b_n$) for all n , so both expansions of our α are infinite and the recursion is the one of the papers.
- `EventuallyPeriodic b` means $\exists N, k$ with $k > 0$ and $b(n + k) = b(n)$ for all $n \geq N$. `PadicBounded p b` means $\exists C \forall n, |b_n|_p \leq C$.

3 The counterexample

Lemma 3 (`exists_beta`). *There is $\beta \in \mathbb{Q}_p$ with $p\beta^2 + \beta - p = 0$ and $|\beta|_p < 1$.*

Proof. Apply Hensel’s lemma (`Mathlib.hensels_lemma`) to $F = pX^2 + X - p \in \mathbb{Z}[X]$ at $a = 0$. We have $|F(0)|_p = |p|_p = 1/p < 1 = |F'(0)|_p^2$, since $F'(0) = 1$. This gives a root $z \in \mathbb{Z}_p$ with $|z - 0|_p < 1$. $\square$

Note $\beta \neq 0$, since $F(0) = -p \neq 0$. Put $\alpha = \beta^{-1}$. Dividing $p\beta^2 + \beta - p = 0$ by $p\beta$ gives

$$\alpha - \frac{1}{p} = \beta, \quad p\alpha^2 - \alpha - p = 0 \quad (\text{alpha_sub, alpha_eq}). \quad (1)$$

Lemma 4 (`alpha_algebraic`, `alpha_irrational`). *α is algebraic over $\mathbb{Q}$, and $\alpha \notin \mathbb{Q}$.*

Proof. α is a root of the nonzero polynomial $pX^2 - X - p \in \mathbb{Q}[X]$. Suppose $\alpha = r \in \mathbb{Q}$. Then $pr^2 - r - p = 0$, so $(2pr - 1)^2 = 4p^2 + 1$, and $4p^2 + 1$ is the square of a rational number. By `Rat.isSquare_natCast_iff` it is then the square of a natural number m . But $(2p)^2 < 4p^2 + 1 < (2p + 1)^2$, a contradiction. $\square$

Theorem 5 (Browkin I; `bcfCQ_alpha`, `bcfPQ_alpha`). *For all n : $\alpha_n = \alpha$ and $a_n = 1/p$. So the Browkin expansion is $[1/p; 1/p, 1/p, \dots]$.*

Proof. $1/p \in J_p$ because $p \geq 3$. Also $|\alpha - 1/p|_p = |\beta|_p < 1$ by (1). By uniqueness, $s(\alpha) = 1/p$, and then $\alpha_1 = (\alpha - 1/p)^{-1} = \beta^{-1} = \alpha$. Now induct on n . Finally, $\alpha_n - a_n = \beta \neq 0$, so the algorithm never stops. $\square$

Theorem 6 (Browkin II; `bcf2_first_steps`, `bcf2CQ_periodic`). *Let $w = (1 + \beta)^{-1}$. The complete quotients of Browkin II are $\alpha, \alpha, w, -\alpha - 1, -w, \alpha + 1, w, \dots$, with $\alpha_{n+4} = \alpha_n$ for all $n \geq 2$. The partial quotients are $1/p, 1/p - 1, 1, -1/p, -1, 1/p$.*

Proof. We have $|1 + \beta|_p = 1$ (ultrametric inequality) and $w - 1 = -\beta w$, so $|w - 1|_p < 1$. Each step is certified by the uniqueness of s or t .

- $n = 0$ (even): $s(\alpha) = 1/p$, so $\alpha_1 = \alpha$.
- $n = 1$ (odd): $t(\alpha) = 1/p$, since $1/p \in K_p$ and $|\alpha - 1/p|_p = |\beta|_p \leq 1$. Since $|\alpha - t(\alpha)|_p = |\beta|_p < 1$, we get $v_p \neq 0$, so $b_1 = 1/p - 1$ and $\alpha_2 = (1 + \beta)^{-1} = w$.
- $n = 2$: $s(w) = 1$, so $\alpha_3 = (w - 1)^{-1} = -\alpha - 1$.
- $n = 3$: $-\alpha - 1 + 1/p = -(1 + \beta)$ has norm 1. So $t = -1/p$, $b_3 = -1/p$ and $\alpha_4 = -w$.
- $n = 4$: $s(-w) = -1$, so $\alpha_5 = (1 - w)^{-1} = \alpha + 1$.
- $n = 5$: $\alpha + 1 - 1/p = 1 + \beta$ has norm 1. So $b_5 = t = 1/p$ and $\alpha_6 = w = \alpha_2$.

The rule for step n depends only on the parity of n (`browkin2Digit_add_four`). So $\alpha_6 = \alpha_2$ propagates to $\alpha_{n+4} = \alpha_n$ for all $n \geq 2$, and the same holds for b_n . All complete quotients are nonzero (`bcf2CQ_ne_zero`), so $\alpha_n \neq b_n$ and the algorithm never stops. $\square$

An eventually periodic sequence has finite range (`EventuallyPeriodic.finite_range`), and a sequence of rationals with finite range is p -adically bounded (`padicBounded_of_finite`). Here the bound is $|b_n|_p \leq p$.

Theorem 7 (`conjecture702_false`). *For every odd prime p , none of the following four statements holds.*

- *ClauseNonPeriodicI p : every irrational algebraic $\alpha \in \mathbb{Q}_p$ has a Browkin I expansion that is not eventually periodic.*
- *ClauseUnboundedI p : every irrational algebraic $\alpha \in \mathbb{Q}_p$ has Browkin I partial quotients that are not p -adically bounded.*
- *ClauseNonPeriodicII p and ClauseUnboundedII p : the same two statements for Browkin II.*

The witness is α above. All the facts about it are collected in `browkin_counterexample`.

4 Lean formalization and axiom audit

The file is `lean/Conjecture702/Basic.lean` (565 lines; only import `Mathlib`). Its main results are:

- `theorem C702.conjecture702_false (hp2 : p ≠ 2) : ¬ ClauseNonPeriodicI p ∧ ¬ ClauseUnboundedI p ∧ ¬ ClauseNonPeriodicII p ∧ ¬ ClauseUnboundedII p`;
- `C702.browkin_counterexample`. It gives an α with $p\alpha^2 - \alpha - p = 0$, algebraic, irrational, both expansions non-terminating, Browkin I constant equal to $1/p$, and Browkin II periodic from index 2 with period 4. Both are eventually periodic, with finite range and p -adically bounded;

- `C702.browkinS_spec`, `C702.browkinT_spec`: s and t exist for odd p .

lake build succeeds. `#print axioms` reports `[propext, Classical.choice, Quot.sound]` for each of these theorems. There is no `sorry` and no `native_decide`.

References

- [CMT] L. Capuano, N. Murru, L. Terracini, *On periodicity of p -adic Browkin continued fractions*, arXiv:2010.07364, <https://arxiv.org/abs/2010.07364>.
- [MRS] N. Murru, G. Romeo, G. Santilli, *On the periodicity of an algorithm for p -adic continued fractions*, arXiv:2201.12019, <https://arxiv.org/abs/2201.12019>.